Tech Crunch Assignment

Jeff Hackmeister

Northeastern University

ALY6030 Data Warehousing & SQL

Professor Nayem Rahman

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Part 1 – Normalization

1. *What is a good choice for a primary key here? In contrast, give an example of an attribute (or composite) that would **not** be a valid primary key.*

    a. The best option for a primary key for the tech crunch dataset would be the fund_id variable. It provides a unique value for each row of the table and is a simple integer value which is easy to continue providing as the table grows.

2. *Does the table satisfy 1NF? Why or why not?*

    a. Yes, this table satisfies 1NF. Each value in the table is unique and only contains atomic values and there are no repeating groups present.

3. *Does the table satisfy 2NF? Why or why not?*

    a. No, the table does not satisfy 2NF. The company value is repeated and the numEmps, city, and state values are dependent on those values.

4. *Does the table satisfy 3NF? Why or why not?*

    a. Since the table does not satisfy 2NF, it cannot satisfy 3NF. To meet 3NF, a second table should be created with company information and a primary key/foreign key relationship between the company table and the remaining funding table – which would need a new primary key.

5. *Sketch a proposed Entity-Relationship diagram that would bring this dataset into 3NF.*

Company		Funding
company_id (PK)	1 <---- M	company_id (FK)
company_name		funding_id (PK)
category		fundedDate
numEmps		raisedAmt
city		raisedCurreny
state		round

Part 2 – Case Study

1. *Use the INSERT INTO statement to insert new information about two (2) completely new recipes of your choosing into the database.*

```
INSERT INTO `categories` VALUES (3,'Breakfast');

INSERT INTO `ingredients` VALUES (20, 'Black Beans'), (21, 'Power Greens'), (22, 'Light Brown Sugar'), (23,  'Chocolate Chips');

INSERT INTO `recipe_main` VALUES (3,'High Fiber Breakfast',1,'Eggs and beans',5,5,1,1,'Add beans to skillet to warm. Add two eggs and season with salt. Top wit
(4,'Chocolate Chip Cookies',2,'Classic cookies',30,15,30,2,'Whisk together the flour, baking soda and 1 teaspoon salt in a large bowl. Beat the butter and both

UPDATE recipe_main
SET category_id = 3
WHERE recipe_id = 3;

INSERT INTO `rec_ingredients` VALUES (17, 3, 2.00, 20), (18, 3, 2, 21), (19, 3, 2, 12), (20, 3, 2, 15), (21, 4, 1.25, 2), (22, 4, 1, 16), (23, 4, 12, 17),
(24, 4, 0.75, 22), (25, 4, 2, 12), (26, 4, 1, 13), (27, 4, 12, 23) ;
```

2. *Write only one SQL query that returns all information on only the two new recipes you inserted from all the tables you created in step 1 above.*

```
SELECT
    recipe_main.recipe_id,
    recipe_main.rec_title,
    recipe_main.recipe_desc,
    recipe_main.prep_time,
    recipe_main.cook_time,
    recipe_main.servings,
    recipe_main.difficulty,
    recipe_main.directions,
    categories.category_name,
    ingredients.ingred_name
FROM recipe_main
JOIN categories ON recipe_main.category_id = categories.category_id
JOIN rec_ingredients ON recipe_main.recipe_id = rec_ingredients.recipe_id
JOIN ingredients ON rec_ingredients.ingredient_id = ingredients.ingredient_id
WHERE recipe_main.rec_title IN ('High Fiber Breakfast', 'Chocolate Chip Cookies');
```

recipe_id	rec_title	recipe_desc	prep_time	cook_time	servings	difficulty	directions	category_name	ingred_name
3	High Fiber Breakfast	Eggs and beans	5	5	1	1	Add beans to skillet to warm. Add two eggs and...	Breakfast	Black Beans
3	High Fiber Breakfast	Eggs and beans	5	5	1	1	Add beans to skillet to warm. Add two eggs and...	Breakfast	Power Greens
3	High Fiber Breakfast	Eggs and beans	5	5	1	1	Add beans to skillet to warm. Add two eggs and...	Breakfast	Egg(s)
3	High Fiber Breakfast	Eggs and beans	5	5	1	1	Add beans to skillet to warm. Add two eggs and...	Breakfast	Salt
4	Chocolate Chip Cookies	Classic cookies	30	15	30	2	Whisk together the flour, baking soda and 1 tea...	Desserts	Flour
4	Chocolate Chip Cookies	Classic cookies	30	15	30	2	Whisk together the flour, baking soda and 1 tea...	Desserts	Baking powder
4	Chocolate Chip Cookies	Classic cookies	30	15	30	2	Whisk together the flour, baking soda and 1 tea...	Desserts	Butter, softened
4	Chocolate Chip Cookies	Classic cookies	30	15	30	2	Whisk together the flour, baking soda and 1 tea...	Desserts	Light Brown Sugar
4	Chocolate Chip Cookies	Classic cookies	30	15	30	2	Whisk together the flour, baking soda and 1 tea...	Desserts	Egg(s)
4	Chocolate Chip Cookies	Classic cookies	30	15	30	2	Whisk together the flour, baking soda and 1 tea...	Desserts	Vanilla extract
4	Chocolate Chip Cookies	Classic cookies	30	15	30	2	Whisk together the flour, baking soda and 1 tea...	Desserts	Chocolate Chips

3. *Write a SELECT query that identifies the recipe name, category name, and ingredient name, and ingredient amount. No other variables should be included. Your output should be sorted first by descending category name, then by ascending recipe name, followed by descending ingredient name.*

```sql
SELECT
    recipe_main.rec_title AS 'Recipe Name',
    categories.category_name AS 'Category Name',
    ingredients.ingred_name AS 'Ingredient Name',
    rec_ingredients.amount AS 'Ingredient Amount'
FROM recipe_main
LEFT JOIN categories ON recipe_main.category_id = categories.category_id
LEFT JOIN rec_ingredients ON recipe_main.recipe_id = rec_ingredients.recipe_id
LEFT JOIN ingredients ON rec_ingredients.ingredient_id = ingredients.ingredient_id
ORDER BY
    categories.category_name DESC,
    recipe_main.rec_title ASC,
    ingredients.ingred_name DESC;
```

Recipe Name	Category Name	Ingredient Name	Ingredient Amount
Chicken Marsala	Entree	Sliced mushrooms	2.00
Chicken Marsala	Entree	Rosemary, dried and crushed	0.25
Chicken Marsala	Entree	Parsley, minced	2.00
Chicken Marsala	Entree	Parmesan cheese, grated	2.00
Chicken Marsala	Entree	Olive oil	2.00
Chicken Marsala	Entree	Marsala wine	0.75
Chicken Marsala	Entree	Flour	2.00
Chicken Marsala	Entree	Chicken breast halves, boneless	4.00
Chicken Marsala	Entree	Butter	2.00
Absolute Brownies	Desserts	White sugar	1.00
Absolute Brownies	Desserts	Vanilla extract	1.00
Absolute Brownies	Desserts	Unsweetened cocoa powder	0.33
Absolute Brownies	Desserts	Salt	0.25
Absolute Brownies	Desserts	Olive oil	0.50
Absolute Brownies	Desserts	Flour	0.50
Absolute Brownies	Desserts	Egg(s)	2.00
Chocolate Chip C...	Desserts	Vanilla extract	1.00
Chocolate Chip C...	Desserts	Light Brown Sugar	0.75
Chocolate Chip C...	Desserts	Flour	1.25
Chocolate Chip C...	Desserts	Egg(s)	2.00
Chocolate Chip C...	Desserts	Chocolate Chips	12.00
Chocolate Chip C...	Desserts	Butter, softened	12.00
Chocolate Chip C...	Desserts	Baking powder	1.00
High Fiber Breakfast	Breakfast	Salt	2.00
High Fiber Breakfast	Breakfast	Power Greens	2.00
High Fiber Breakfast	Breakfast	Egg(s)	2.00
High Fiber Breakfast	Breakfast	Black Beans	2.00